

Fundamentals of Data Science

Course syllabus · Bachelor's, Term 3

Software	Google Colab only. Nothing to install, nothing to buy
Prerequisites	None. No prior programming experience is assumed
Instructor	
Email	
Office hours	
Course page	sabokrou.github.io/teaching/data-science-fundamentals

About this course

Data science is not a collection of algorithms. It is the practice of answering a question with data, saying how certain the answer is, and explaining it to someone who has to act on it.

This course teaches that practice from the beginning. You will learn to turn a vague concern into a question data can answer, to check what a dataset can and cannot tell you, to clean it honestly, to summarise and chart it, to build simple models, and to say clearly what your result does **not** establish. Machine learning appears in Week 9, and by then you will know why it is not the first thing that matters.

Ethics is not a separate topic bolted on at the end. It runs through the course — framing and consent in Week 1, what cleaning deletes in Week 3, honest charts in Week 6, thresholds and unequal impact in Week 10, and privacy in Week 12.

Everything runs in Google Colab in your browser. If you have never written a line of code, **Session 0 before Week 1 is built for you** and assumes nothing.

Materials

Everything is provided. There is no textbook to buy.

Item	What it is
Volume 1 — Main Text	The textbook. 15 chapters, 45 worked examples, 28 figures
Volume 2 — Reference	Model answers, code cookbook, formula sheet, glossary
Lab Manual	Lab 0 preparation plus the twelve two-hour labs
Lab 0 — Colab and Python	The Python practical, with a starter notebook
data/ (seven CSV files)	Synthetic university data. No real student record is in it
DATA_DICTIONARY.md	What every column means, its units, and its quirks

Put the data folder in your Google Drive as `MyDrive/fds/data/`. The same 1,200 students appear in Week 1 and in Week 12, so effort spent understanding them early keeps paying.

Before the first week

Complete Lab 0 (Preparation) from the Lab Manual. It takes twenty minutes and gets Colab, Drive and the data working. Do it at home, before the first taught session.

You will know it worked when the setup cell prints:

```
(1200, 7) (4665, 5) (16165, 6) (1200, 7)
```

If it does not work, say so in the first session. It will not fix itself, and every lab from Week 1 assumes it works.

Schedule

Wk	Lecture	Ch	Lab
0	Python for Data Science — for students new to programming	—	Lab 0 — Colab and Python
1	Framing questions; the claim ladder; the lifecycle; sources, sampling frames and proxies	1, 2	Lab 1 — Rows, joins, and what is missing
2	Centre, spread, shape, standardisation	3	Lab 2 — Summaries by hand, then in code
3	Quality dimensions; kinds of missingness; duplicates; outliers	4	Lab 3 — Clean the registry export
4	Scaling, encoding, date features; leakage; pipelines	5	Lab 4 — Pipelines and leakage
5	Structured exploration; confounding; Simpson's paradox	6	Lab 5 — Exploratory analysis that survives scrutiny
6	Visual encoding; baselines; colour and accessibility; message titles	7	Lab 6 — Chart critique and redesign
7	Probability; conditional direction; Bayes with natural frequencies	8	Lab 7 — Probability and simulation
8	Sampling variability; confidence intervals; tests; effect size; A/B design	9	Lab 8 — Intervals, resampling and power
9	Baselines; linear regression; metrics; residuals; cross-validation	10	Lab 9 — The regression workflow
10	Logistic regression; k-NN and trees; confusion matrix; thresholds; fairness	11, 12	Lab 10 — Classification, end to end
11	Distance; k-means; choosing k; PCA; naming responsibly	13	Lab 11 — Clustering with honest uncertainty
12	Time series and text; dashboards; privacy; contestability	14, 15	Lab 12 — Time series, text and a dashboard

The midterm exam falls in Week 6. The exact slot will be confirmed in class.

Assessment

Component	Due	Weight
Midterm exam	Week 6	20%
Final exam	Exam period	25%
Weekly lab notebooks (best 9 of 12)	Weekly	15%
Data Quality Report	Week 4	10%
EDA and Visualization Brief	Week 7	10%
Modelling Assignment	Week 11	10%
Capstone project and dashboard	Week 12	10%

What you actually hand in

Exams (45%).

Written, in person. Every question shows you something — a table, a printed result, five lines of code and their output, a chart with a title — and asks a judgement question about it. **No question asks you to define a term.** Memorising the glossary will not pass either paper; doing the labs will. The midterm covers Weeks 1–5. The final covers the whole course, weighted toward interpretation.

Weekly lab notebooks (15%).

One notebook each week, submitted in the LMS. These are not finely marked: you get pass, revise, or fail. “Revise” means fix it and send it back. Only your best 9 of 12 count, so one bad week costs you nothing.

Data Quality Report (10%, Week 4).

A notebook that finds what is wrong with the data — missing values, impossible values, duplicates, columns that mean something other than their name — plus a short report saying what you did about each problem and why.

EDA and Visualization Brief (10%, Week 7).

A short document of charts. Each chart needs a title that states its point, not just its contents, and you name what kind of claim it supports.

Modelling Assignment (10%, Week 11).

A notebook with a model. But the model is not where the marks are: **25 of the 100 marks are for justifying where you set the threshold** — at what score you flag someone, who gets wrongly flagged, and who gets missed.

Capstone project and dashboard (10%, Week 12).

You choose your own public dataset — not the course one — and do the whole job: question, data, cleaning, analysis, charts, dashboard. **20 of the 100 marks are for the limitations section:** what your work cannot tell you, and who could be harmed if you are wrong.

The two rules

These apply to everything you submit, and they are enforced in marking from Week 4.

1. Runtime → Restart and run all, before you submit anything.

Your notebook must run correctly from top to bottom from a clean start. If it only works because you ran the cells in a particular order, it does not count. An analysis nobody can reproduce cannot be checked — including by you.

2. A number without a sentence interpreting it is not an answer.

Every result needs a sentence saying what it means, and at least one limitation. This course does not mark whether you got an answer. It marks whether you can say what the answer means, and what it does not mean.

One last thing

The habit this whole course rests on is a single sentence, and you will meet it in the first lecture:

Before any count, mean, groupby or merge — state what one row of the table is.

It sounds trivial. It is the difference between an answer that is right and an answer that is merely computed.

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